

1. Simulation process. Initialize with a random distribution of a pair of average concentration and internal energy $(\bar{c}, \bar{E})$.

2. A coupled partial melting Stokes-Darcy system. Strong form

$$\begin{aligned}
 (2.1) \quad & \tilde{\mathbf{v}}_r + \frac{k_0 \phi_f^{1+\Theta}}{\mu_f} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
 & \mu_s \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0 \\
 & \nabla q - \nabla \cdot \hat{\sigma}(\mathbf{v}_s) = -(1 - \phi_f) \rho_r \mathbf{g} \\
 & \mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} q) = 0
 \end{aligned}$$

List SI units

q = pascal.

$\mathbf{v}$ = m/s

Non dimensionalization choice

- $\tilde{\mathbf{v}} = \frac{\mathbf{v}_r}{k_0 \rho_r g_y / \mu_f}$
- $\tilde{q} = \frac{q}{\rho_r g_y l_0}$
- $\tilde{x} = \frac{x}{l_0}$
- $l_0 = \left(\frac{k_0 \mu_s}{\mu_f} \right)^{1/2}$

The resulting non dimensionalized form is therefore

$$\begin{aligned}
 (2.2) \quad & \tilde{\mathbf{v}}_r + \phi_f^{1+\Theta} \nabla (\phi_f^{-1/2} \tilde{q}_f) = 0 \\
 & \phi_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} \tilde{q}) = 0 \\
 & \nabla \tilde{q} - \nabla \cdot \hat{\sigma}(\tilde{\mathbf{v}}_s) = -(1 - \phi_f) \rho_r \hat{\mathbf{g}} \\
 & \nabla \cdot \tilde{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1 - \phi_f} (\tilde{q}_f - \phi_f^{1/2} \tilde{q}) = 0
 \end{aligned}$$

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3} \nabla \cdot \mathbf{v} \mathbf{I}$

$$\begin{aligned}
 (2.3) \quad & (2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\tilde{q}_h, \nabla \cdot \Psi_s) = -(\phi_s \hat{\mathbf{f}}, \Psi_s) \\
 & (\nabla \cdot \tilde{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s} \tilde{q}_h, \omega \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \tilde{q}_{f,h}, \omega \right) = 0 \\
 & (\tilde{\mathbf{v}}_{r,h}, \Psi_r) - (\tilde{q}_{f,h}, \hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r)) = 0 \\
 & (\hat{\phi}_f^{-1/2} \nabla \cdot (\phi_f^{1+\Theta} \tilde{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s} \tilde{q}_{f,h}, \omega_f \right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s} \tilde{q}_h, \omega_f \right) = 0
 \end{aligned}$$

2.1. Variational form and discretization.

2.1.1. Stokes and variational formulation.

$$(2.4) \quad \begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(2.5) \quad \begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i} \\ &+ \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $\langle \mathbf{s}, \mathbf{v} \rangle_{\Gamma_i}$. Traditionally a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0 \text{ on } \Gamma_u\}$ is employed here.

$$(2.6) \quad \begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

2.2. Darcy.

$$(2.7) \quad \begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned}$$

weak form

$$(2.8) \quad \begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned}$$

Test space $\{\mathbf{v} \in H(\text{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

2.2.1. A degenerate Darcy test problem.

$$(2.9) \quad \begin{aligned} \mathbf{v} + \phi \nabla(\hat{\phi}^{-1/2} p) &= \mathbf{f} \\ \hat{\phi}^{-1/2} \nabla \cdot (\phi \mathbf{v}) &= g \end{aligned}$$

We manufacture a solution with degenerate porosity in the target physical domain $x, y \in [-1, 1]$.

$$(2.10) \quad \phi = \begin{cases} \cos^m(n(x^2 + y^2)) & x^2 + y^2 < \pi/2n \\ 0 & \text{otherwise} \end{cases}$$

where m is an even number, and $\pi/2n < 1$ another choice of test case is shown as follows:

$$(2.11) \quad \phi = \begin{cases} 0 & x \leq 0 \\ \phi_+ x^4 & x > 0 \end{cases}$$

Define an arbitrary velocity and pressure

$$(2.12) \quad \begin{aligned} \mathbf{v} &= [\frac{x}{5}, -y]^T \\ p &= \hat{\phi}^{1/2} \end{aligned}$$

The source term is calculated as follows:

$$(2.13) \quad f = \mathbf{v}$$

and the divergence term is:

$$(2.14) \quad \begin{aligned} g &= \phi^{-1/2} \phi_+ \nabla \cdot [\frac{x^5}{5}, -x^4 y]^T \\ &= \phi^{-1/2} \phi_+ (x^4 - x^4) = 0 \end{aligned}$$

a divergence free test case

2.2.2. Modified BR element. The Bernardi-Raugel element has been researched extensively in previous literature [2]. We state that our new function space satisfies the following hypotheses:

- For any q in $H^m(\Omega)$

3. Linear system and Uzawa type algorithm. Boundary conditions are already included in the linear system formulation. Original linear system:

$$(3.1) \quad \begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}$$

This is a double saddle point system. Symmetrically permute original linear system and form an enlarged saddle point system.

$$(3.2) \quad \begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}$$

$$(3.3) \quad \mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix} \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix} \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \end{bmatrix} \quad \mathbf{G} = \begin{bmatrix} -\mathbf{f}_d \\ -\mathbf{g}_d \end{bmatrix}$$

In this newly formed system, the $\mathbf{A}$ matrix maintains as a positive definite matrix. we solve this typical saddle point system with uzawa iteration. Uzawa algorithm has been discussed extensively at [3], [4].

Consider the linear system of equation:

$$(3.4) \quad \begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}$$

Algorithm 3.1 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \mathbf{Q}^{-1}(\mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G})$
-

In the previous literature [4], $\mathbf{Q}$ is approximated with $\tau\mathbf{I}$, which works well with the stokes part. However, it converges very slow with the same approximation when applied to the Darcy part. We need a better preconditioner approximating inverse of the corresponding schur complement. We use MINRES algorithm [1] due to schur complement being symmetric and indefinite.

Algorithm 3.2 Preconditioned Inexact Uzawa Iteration

- 1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
 - 2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
 - 3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
 - 4: Obtain $\tilde{\mathbf{y}} = \operatorname{argmin}_{\mathbf{y} \in V} \|(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}\|$
 - 5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$
-

Based on the above algorithm, we proposed a new block preconditioner for our coupled linear system.

$$(3.5) \quad \mathbf{Q}^{-1} = \begin{bmatrix} \mathbf{S}_s^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d^{-1} \end{bmatrix}$$

where the schur complement matrices are approximated with MINRES method.

3.1. Convergence and stability analysis. Uzawa iteration is in fact a fixed point iteration procedure. Usually, uzawa iteration convergences slowly.

3.1.1. Anderson acceleration. Accelerating uzawa iteration has been discussed in the previously literatures. Anderson acceleration (Anderson mixing) [6]

4. Test cases. We test the linear system a little bit different from the linear system above.

4.1. Constant porosity and balanced pressure (divergence free velocity field).

4.2. Const porosity and unbalanced pressure.

$$(4.1) \quad \check{q} = 0 \quad \check{q}_f = 2(x+y) \quad \check{\mathbf{v}}_r = -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} \quad \check{\mathbf{v}}_s = \frac{\phi_f^{1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix}$$

$$(4.2) \quad \nabla \cdot \check{\mathbf{v}}_r = \frac{-2\phi_f^{-1/2}}{1-\phi_f}(x+y)$$

$$(4.3) \quad \mathcal{D}\check{\mathbf{v}}_s = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x & 0 \\ 0 & y \end{bmatrix} \quad (\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I} = \frac{2\phi_f^{1/2}}{1-\phi_f} \begin{bmatrix} x+y & 0 \\ 0 & x+y \end{bmatrix}$$

$$(4.4) \quad \begin{aligned} \check{\mathbf{v}}_r + \phi_f^{1/2} \nabla \check{q}_f &= -\frac{\phi_f^{-1/2}}{1-\phi_f} \begin{pmatrix} x^2 \\ y^2 \end{pmatrix} + \phi_f^{1/2} \begin{pmatrix} 2 \\ 2 \end{pmatrix} \\ \phi_f^{1/2} \nabla \cdot (\check{\mathbf{v}}_r) + \frac{1}{1-\phi_f} \check{q}_f &= \frac{-2}{1-\phi_f}(x+y) + \frac{1}{1-\phi_f} 2(x+y) = 0 \\ \nabla \cdot \check{\mathbf{v}}_s - \frac{\phi_f^{1/2}}{1-\phi_f} \check{q}_f &= \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) - \frac{2\phi_f^{1/2}}{1-\phi_f}(x+y) = 0 \\ -2(1-\phi_f) \nabla \cdot \sigma(\check{\mathbf{v}}_s) &= -2(1-\phi_f) \nabla \cdot (\mathcal{D}\check{\mathbf{v}}_s - \frac{1}{3}(\nabla \cdot \check{\mathbf{v}}_s)\mathbf{I}) = -\frac{8\phi_f^{1/2}}{3} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \end{aligned}$$

5. Hyperbolic Equation System. In the simulatin of partial melting system, there arises multidimensional hyperbolic equations systems.

$$\mathbf{U}_t + \nabla \cdot [F(\mathbf{U}) - A(\mathbf{U}\nabla\mathbf{U})] = 0$$

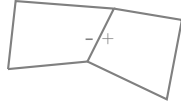
$\mathbf{U}$ consists of two variables, average concentration of one species $\bar{c}_1$ and average total internal energy $\bar{E}$.

$$\bar{c}_t + \nabla \cdot (\mathbf{v}_e \bar{c} - \kappa D \nabla \bar{c}) = 0$$

The finite volume discretization can be written as:

empty

where $\kappa = \frac{\phi_f c_f}{\bar{c}}$ need definition on edge shared between two cells.



There are two κ values defined on both sides of the shared edge.

$$\kappa^\pm = \frac{\phi_f c_f^\pm}{\bar{c}^\pm}$$

where ϕ_f has definition on the edge.

6. Eutectic phase behavior. Mid-ocean ridge forms at the place where temperature is generally lower. The binary eutectic phase behavior describes the chemical behavior of two immiscible crystals from a completely miscible solution.

On a typical eutectic phase behavior diagram, we can locate any phase situation using temperature and composition information. We follows phase behaviors details described in [5]. We made some reasonable adjustments for our mantle system.

There are few key elements that allow precise calculation of phase region and corresponding volumetric fractions.

- Eutectic melting mass fraction is only related to temperature and pressure. Fluid formed by eutectic melting has a constant composition. $\chi_e(T, P)$
- Melting temperature related to pressure. Pressure release is one of the main reason why melting happens in mantle. $T_1(P)$ and $T_2(P)$.
- All melting are eutectic melting. Temperature not exceeds T_1 or T_2 . Therefore density of melting is a function of χ_e .
- We focus on partial melting situation. No full melting happens at our simulation. Hence we have a constant fluid density related to eutectic mass compositon.

We define following dimensionless bulk variables following [5]. We denote volumetric fraction of three phases as olivine ϕ_1 , opx ϕ_2 and melt ϕ_f . We are tracking opx in the mixture for simulation. Stefan number is defined as the ratio of the sensible heat to latent heat, $Ste = \frac{c_p \Delta T}{L}$. We define dimensionless temperature with eutectic temperature as $\mathcal{T} = \frac{T - T_e}{T_{MAX} - T_e}$, and dimensionless pressure as $\mathcal{P} = \frac{P - P_0}{P_0}$, where P_0 represents standard atmospheric pressure. We use melting temperature of olivine at standard atmospheric pressure as 2053 K, and 1753 K for orthopyroxene (opx).

$$(6.1) \quad \begin{aligned} C &= \phi_2 \rho_2 + \phi_f \rho_f \chi_e(\mathcal{T}, \mathcal{P}) \\ H &= \phi_1 \rho_1 c_{p,1} \mathcal{T} + \phi_2 \rho_2 c_{p,2} \mathcal{T} + \phi_f \rho_f (L_f + c_{p,f} \mathcal{T}) \end{aligned}$$

Where the melting latent heat can be affected by multiple things. We can approximate the latent heat of the mixture as

$$L_f = L_1(1 - \chi(\mathcal{T}, \mathcal{P})) + L_2 \chi(\mathcal{T}, \mathcal{P})$$

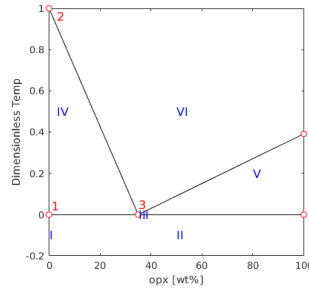
. Furthermore, we can approximate the latent heat as constant

$$\mathcal{L}_f = L_1(1 - \chi_e) + L_2 \chi_e$$

, We can nondimensionalize bulk composition and dimensionless bulk enthalpy as:

$$(6.2) \quad \begin{aligned} \mathcal{C} &= C / \rho_1 \\ \mathcal{H} &= H / (c_{p,1} \rho_1) \end{aligned}$$

The eutectic phase diagram can be simplified as:



6.1. Phase split calculation. In mantle, we mostly have olivine and orthopyroxene (opx) mixture in the rock. Like eutectic melting system e.g. salt and ice, the presence of one phase would largely lower the melting temperature of another phase to the eutectic melting point. We now decompose different phase regions in details. We are tracking bulk concentration of orthopyroxene (opx) in our calculation. We identify following 6 regions with different phase assemblages:

- I Single-phase sub-solidus : olivine
- II Two-phase sub-solidus: olivine + opx
- III Three-phase eutectic: olivine + opx + melt
- IV Two-phase super-eutectic : olivine + melt
- V Two-phase super-eutectic : opx + melt
- VI Single-phase super-liquidus : melt

6.1.1. Region I. Pure olivine. The conditions to be in region I are

$$(6.3) \quad \mathbf{C} = 0 \quad \mathbf{H} \leq 1$$

Hence, the volumetric fraction and temperature can be calculated simply as:

$$(6.4) \quad \phi_1 = 1, \quad \phi_2 = \phi_f = 0 \quad \text{and} \quad \mathcal{T} = \mathcal{H}$$

6.1.2. Region II. Two-phase sub-solidus: olivine + opx. The most common

6.1.3. Region III. In this three-phase eutectic region, all three phases are present. The temperature is fixed at the eutectic point.

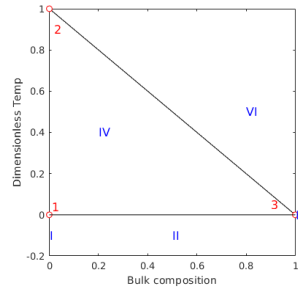
6.1.4. Region IV. In this region, melting starts at the point left to the eutectic melting point.

6.1.5. Region V. In this region, melting starts at the point right to the eutectic melting point.

6.1.6. Region VI. In this region, only melting is present. Hence the volume fraction is given:

$$(6.5) \quad \phi_{oli} = \phi_{opx} = 0, \quad \phi_{melt} = 1$$

6.2. Eutectic melting. In practical, we are focusing on eutectic melting happening when mass fraction of opx is left to eutectic point. We notice that upper mantle consists of more than 80% of olivine according to [7]. Hence, we firstly restrict our calculation to the left side of eutectic point. The simplified left eutectic phase diagram is now:



REFERENCES

- [1] M. A. S. C. C. Paige, Solution of sparse indefinite systems of linear equations, *SIAM. J. Numer. Anal.*, 12 (1975), pp. 617–629.
- [2] G. R. Christine Bernardi, Analysis of some finite elements for the stokes problem, *Math. of Comput.*, 44 (1985), pp. 71–79.
- [3] H. C. Elman and G. H. Golub, Inexact and preconditioned uzawa algorithms for saddle point problems, *SIAM. J. Numer. Anal.*, 31 (1994), pp. 1645–1661.
- [4] A. T. V. James H. Bramble, Joseph E. Pasciak, Analysis of the inexact uzawa algorithm for saddle point problems, *SIAM. J. Numer. Anal.*, 34 (1997), pp. 1072–1092.
- [5] S. D. V. Marc A. Hesse, Jacob S. Jordan and A. V. Oza, Supporting information for "downward oxidant transport through europa's ice shell by density-driven brine percolation", *Geophysical Research Letters*, (2022).
- [6] H. F. W. Nguyenho Ho, Sarah D. Olson, Accelerating the uzawa algorithm, *SIAM. J. Sci. Comput.*, 39 (2017), pp. S461–S476.
- [7] S.-s. S. W.F. McDonough, The composition of the earth, *Chemical Geology*, 120 (1995), pp. 223–253.